

Code Architecture

Obstacle Challenge

For the Obstacle Challenge, the robot will primarily rely on the camera for navigation, with the other sensors serving specific supporting roles.

Sensor Reliance

Sensor	Approx. Reliance	Primary Purpose
Raspberry Pi Camera	75%	Obstacle detection, colour identification, navigation
TOF Sensors	5%	Parking, crash prevention, and final positioning
IMU + Rotary Encoder	Supporting	Parking guidance and path mapping

The TOF sensors will have the highest authority when detecting an immediate risk of collision. The IMU and rotary encoder may also be used after the first round to plot a stable path for the following two rounds.

Code Structure

The selected camera will stream data to the Raspberry Pi Zero 2 W at QQVGA resolution (160 × 120 px), using either HSV or YUV422. The Raspberry Pi Zero 2 W will run either Linux or Raspberry Pi OS Lite.

Core 1 — Computer Vision and Obstacle Detection

Core 1 will handle the initial processing of the camera feed.

1. A lightweight image-processing filter will identify green, red, and purple pixels in the camera feed and determine the colour of the obstacle.
2. Approximately 30% of the upper portion of the camera frame will be disregarded to reduce unnecessary processing and background interference.
3. Once the obstacle colour has been identified, a basic algorithm will determine the obstacle's four corners.

4. The algorithm will initially locate the points with the smallest and largest X and Y coordinates and use basic geometric calculations to determine the remaining two corners.
5. A dedicated noise-filtering system will be implemented to reduce false detections caused by the background and reflections. A luminance check is expected to be used for this purpose.
6. The vision system will be adapted during testing to address problems encountered during real-world operation.
7. Once the obstacle data has been processed, the relevant information will be transmitted to Core 2 for navigation.
8. The camera system will continuously track the detected obstacle corners that have been passed. This information will be used to count laps and trigger the parking manoeuvre.

Core 2 — Navigation and Steering

Core 2 will use the obstacle-corner data generated by Core 1 to guide the robot around the obstacle.

A Pure Pursuit algorithm will be used to determine the desired path around the obstacle. Steering control will primarily be performed by adjusting the servo motor angle.

PWM will be regulated loosely to maintain sufficient control while allowing the robot to operate at an appropriate speed.

Parking Manoeuvre — Preliminary

The following parking sequence is not yet final and will be refined during testing.

After completing the third lap:

1. The robot will immediately reverse.
2. It will position itself at a distance of **X cm** from the wall.
3. **X** will be defined as:
 $X = k \times \text{parking-wall length}$
where **k** is a constant that will be determined through testing and tuning.
4. The robot will then drive alongside the parking area until the TOF sensor on the respective side briefly detects the parking wall.
5. The sensor, positioned approximately at the centre of the robot, will be aligned with the wall.
6. The robot will then use odometry to travel a distance **Y** forward. The value of **Y** will depend on the robot's turning characteristics and will be determined through testing.
7. The robot will reverse into the parking area.
8. Finally, the robot will perform one final turn, assisted by the rear TOF sensor, to align itself parallel to the parking wall.

Open Challenge

The code for the Open Challenge will be considerably simpler.

The robot will primarily:

- Use the camera to determine when turns are required.
- Use the TOF sensors to regulate its position relative to the track and surrounding walls.
- Operate the drive motors at their maximum practical speed.
- Complete parking by stopping a few seconds after crossing the second-last turn of the third lap.

The Open Challenge strategy will therefore prioritize simple camera-based navigation, positional correction through the TOF sensors, and maximum practical motor performance.